



Lab 1

GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

BY

AYESHA MAQSOOD

Section : A

Reg No. : FA24B1-SE-042

Course : Cloud Computing

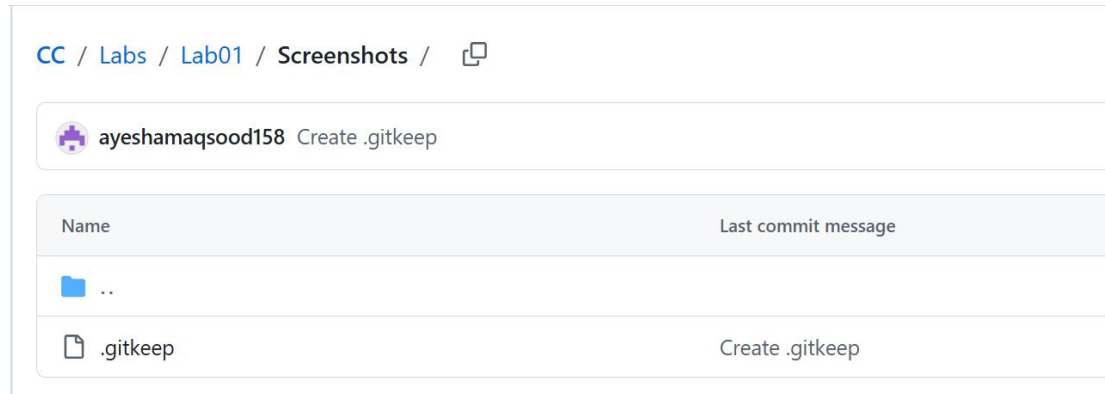
Course code : BSE-410

Github Username : ayeshamaqsood158

Submitted to : Sir Waqas Saleem

Getting started:

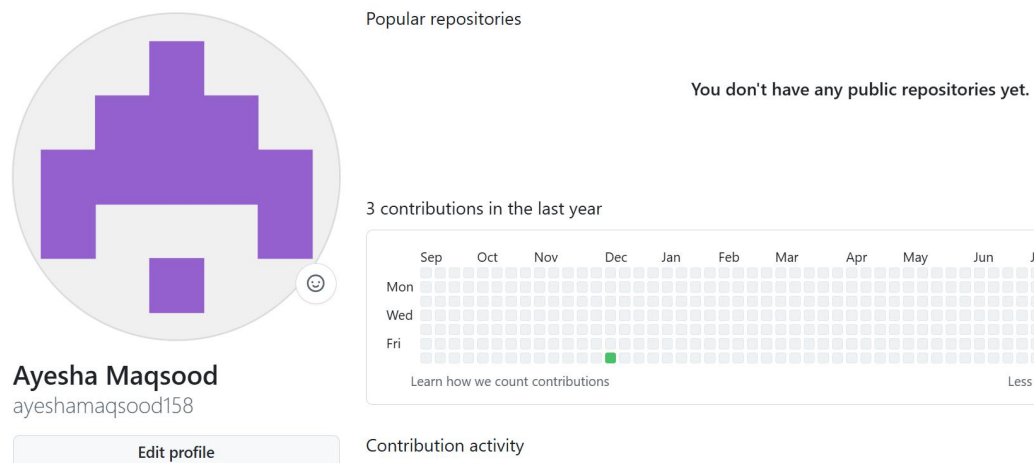
lab01_folder_structure:



The lab01 folder structure created in the GitHub repository: CC / Labs / Lab01 / Screenshots, with a .gitkeep file committed so the empty folder is tracked.

Task 1:

Github_profile:



The newly created GitHub account profile for the username ayeshamaqsood158, showing it is live and publicly visible.

Github_actual_name:

 **Ayesha Maqsood** (ayeshamaqsood158)
Your personal account

Public profile

Account

Appearance

Accessibility

Notifications

Public profile

Name

Ayesha Maqsood

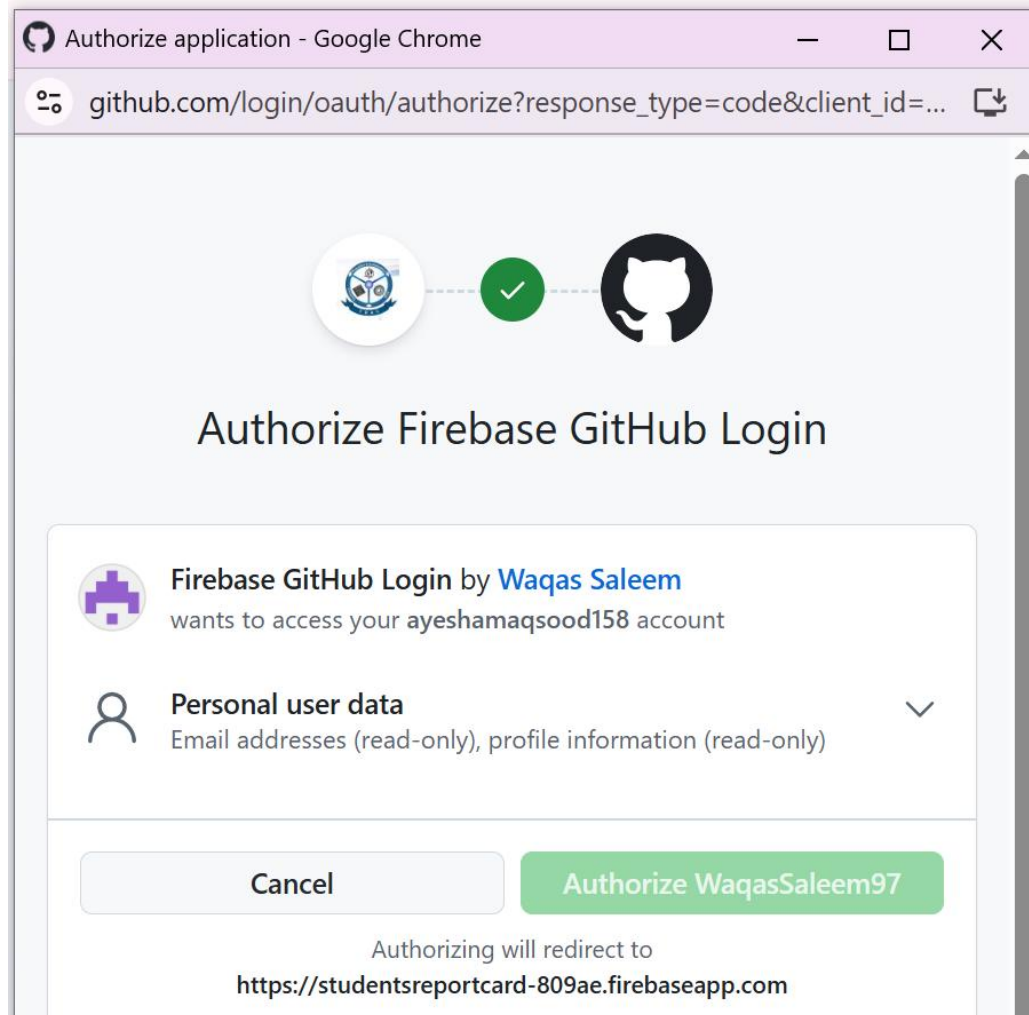
Your name may appear around GitHub where you contribute or are mentioned. You can remove it at any time.

Public email

The GitHub public profile settings with the Name field set to the student's real name, as required by the lab.

Task 2:

Portal_login_page:



The GitHub OAuth authorisation prompt, where the student portal application is granted read-only access to the account to complete sign-in.

Portal_github_profile:

ID

Complete registration

Enter your registration number, course and section.
The enrollment requires approval.

Registration number

2024-BSE-42

Allowed: 2024-BSE-01 to 2024-BSE-99.

Course

SE-665 — Cloud Computing

Section

Section A

Submit enrollment

Sign out

The portal registration form filled in with registration number 2024-BSE-42, course SE-665 Cloud Computing and Section A.

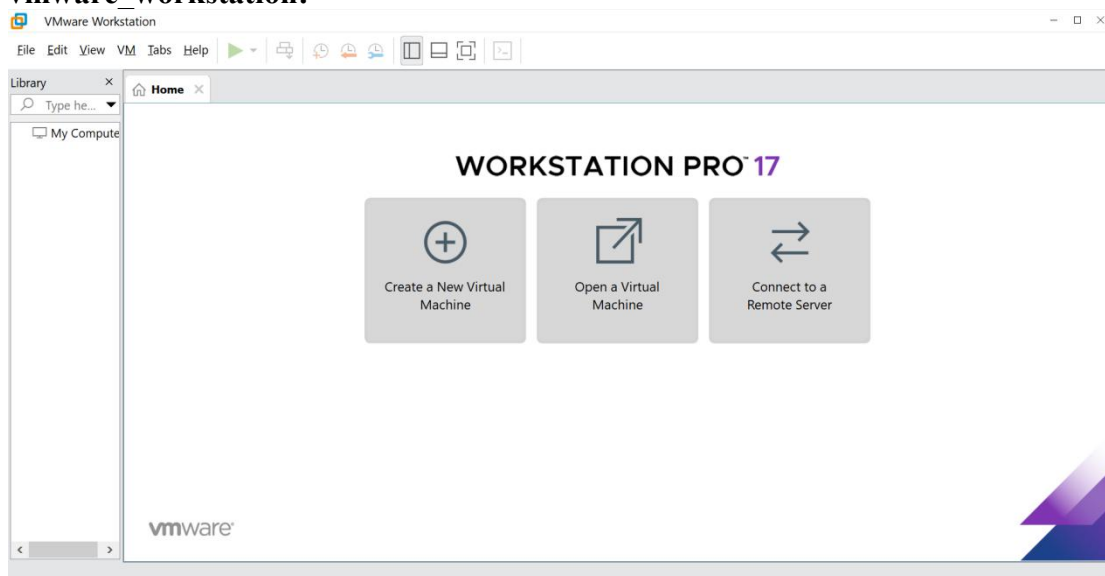
Portal_enrollment_submitted:



The portal's Course Enrollments page listing Cloud Computing (SE-665) Section A with a Pending Approval status, confirming the enrollment request was submitted.

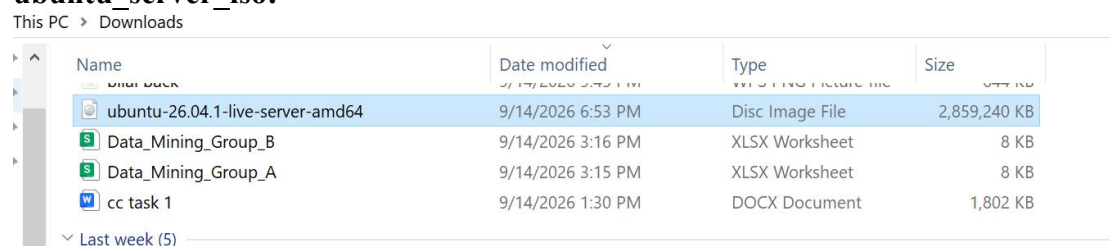
Task 3:

vmware_workstation:



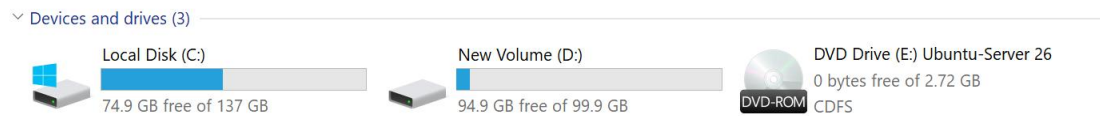
The VMware Workstation Pro 17 home screen, confirming the hypervisor is installed and ready to create a virtual machine.

ubuntu_server_iso:



The downloaded Ubuntu Server image, *ubuntu-26.04.1-live-server-amd64.iso* (about 2.7 GB), saved in the Downloads folder.

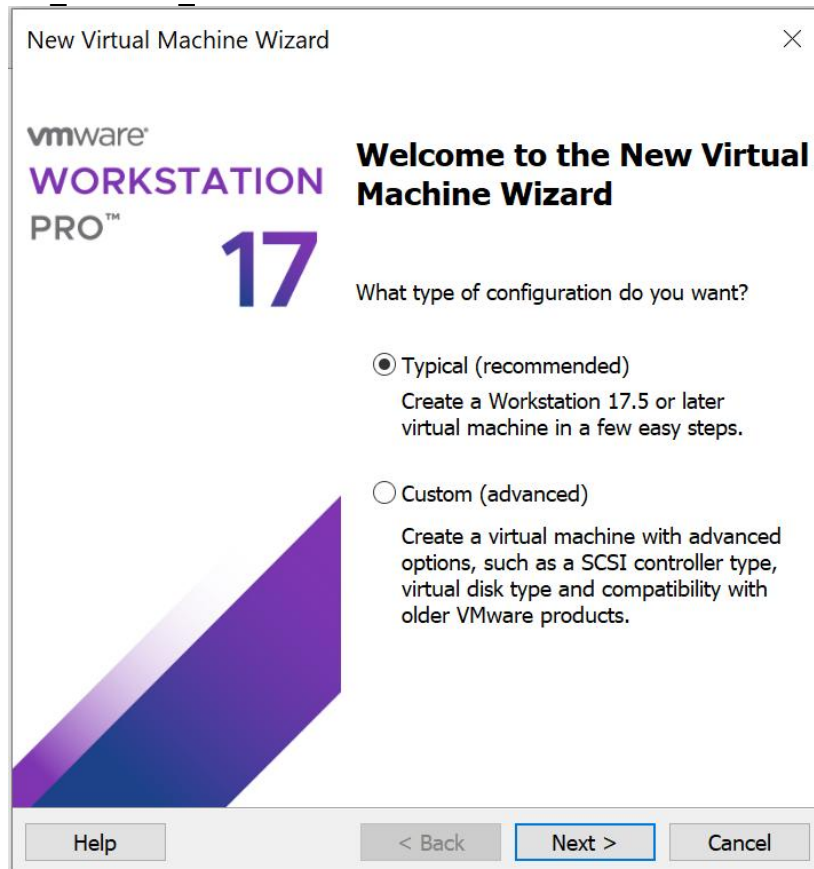
available_storage:



Available disk space on the host machine: 74.9 GB free on drive C: and 94.9 GB free on drive D:, comfortably more than the virtual machine requires.

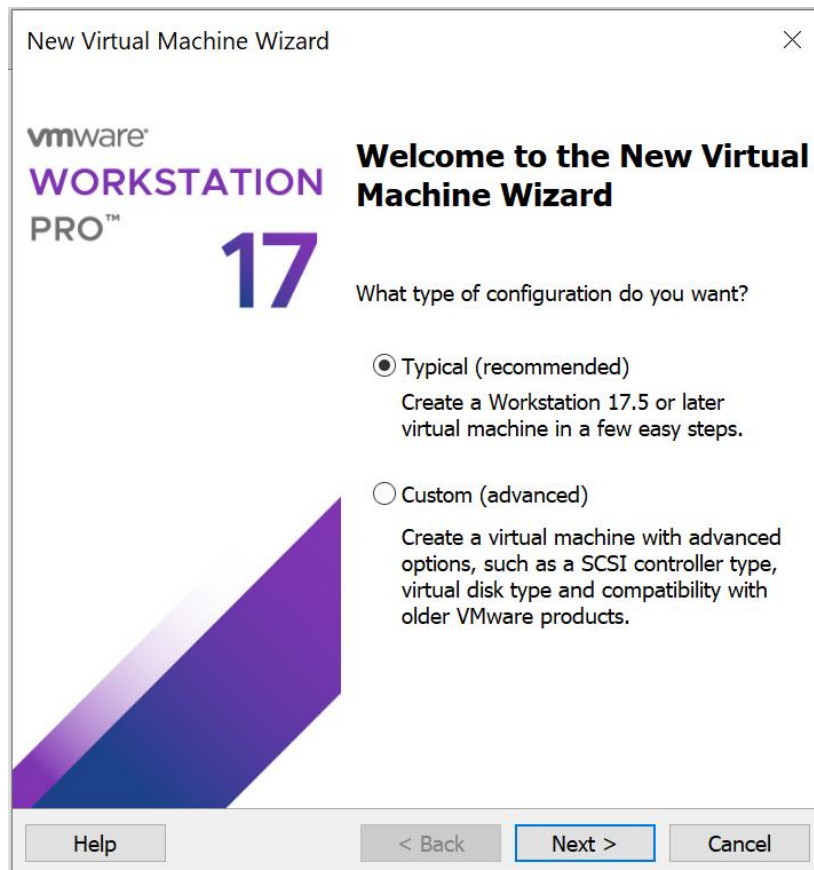
Task 4:

vm_creation_wizard:



The New Virtual Machine Wizard opened in VMware Workstation, the starting point for creating the Ubuntu Server VM.

vm_typical_configuration:



The Typical (recommended) configuration option selected, which creates the VM using default Workstation 17.5 settings.

vm_ubuntu_iso_selected:

New Virtual Machine Wizard

Guest Operating System Installation

A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?

Install from:

☐ Installer disc:

DVD Drive (E:) Ubuntu-Server 26

☒ Installer disc image file (iso):

C:\Users\NEXGEN\Downloads\ubuntu-26.04.1-live-s Browse...

 Ubuntu 64-bit Server 26.04.1 detected.
To use Easy Install, insert the first disc of the set.

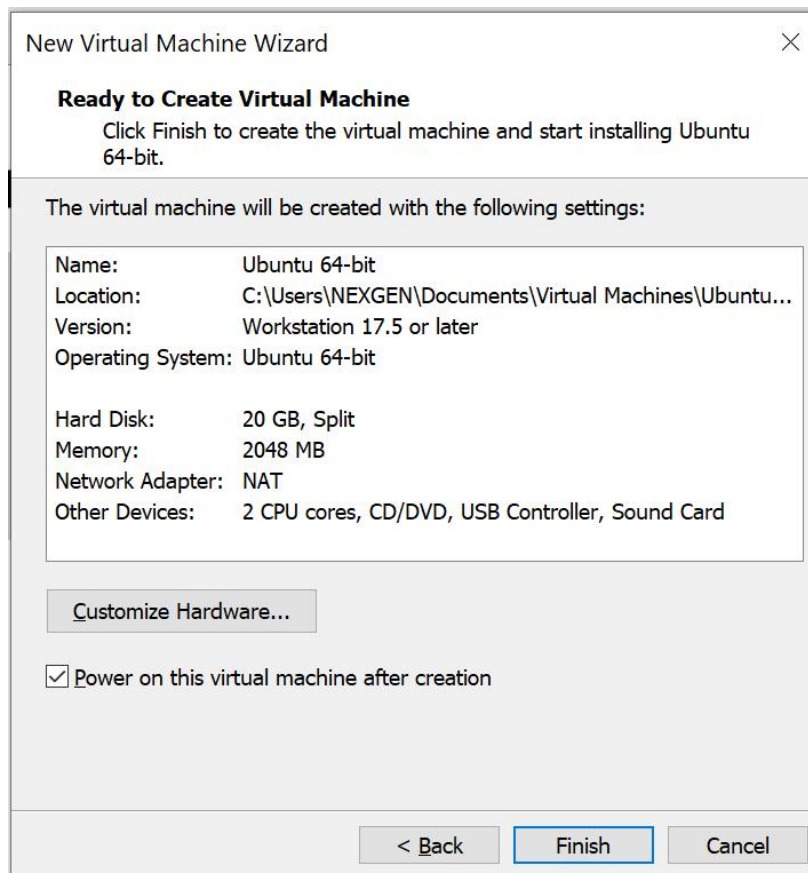
☐ I will install the operating system later.

The virtual machine will be created with a blank hard disk.

Help < Back Next > Cancel

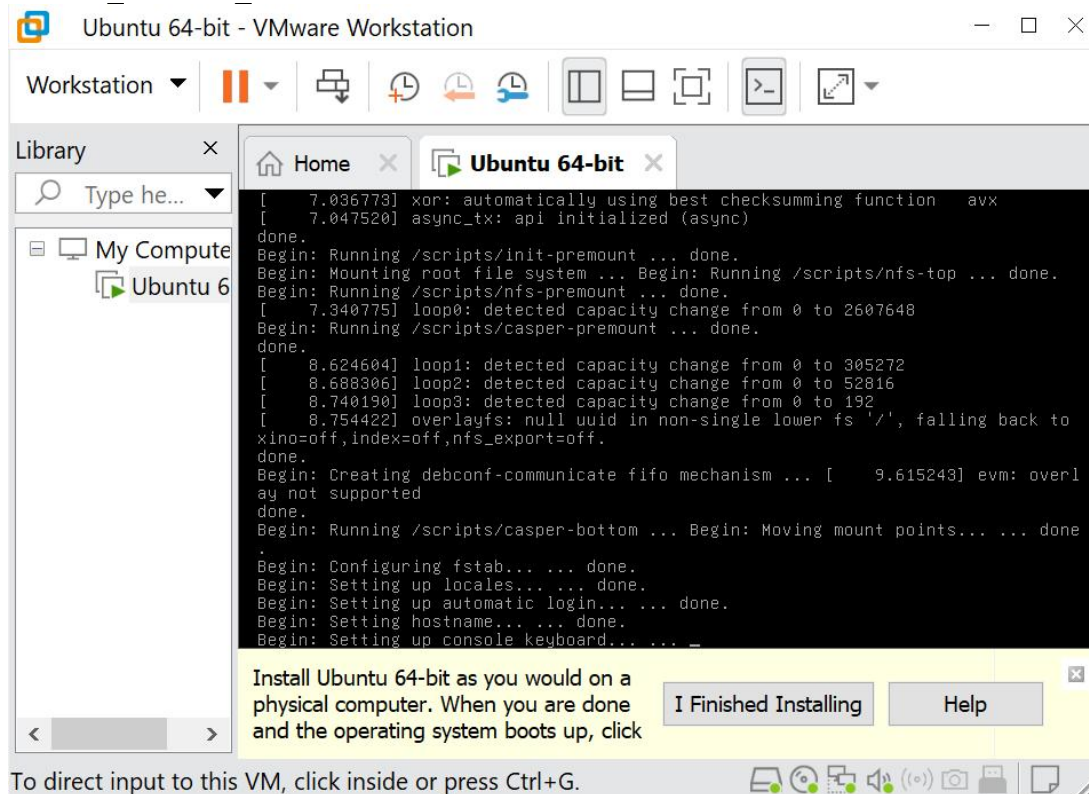
The Ubuntu ISO selected as the installer disc image; VMware detects it automatically as Ubuntu 64-bit Server 26.04.1.

vm_configuration_summary:



The final wizard summary of the virtual machine: Ubuntu 64-bit, 20 GB split virtual disk, 2048 MB memory, 2 CPU cores and a NAT network adapter.

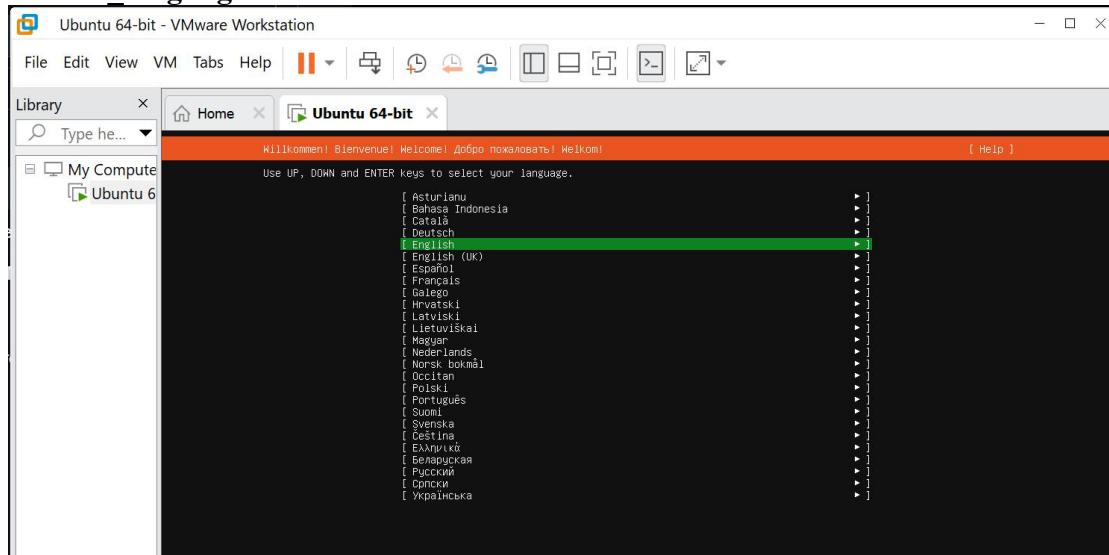
ubuntu_installer_boot:



The virtual machine powered on and booting from the ISO, with the live installer's kernel and start-up messages on screen.

Task 5:

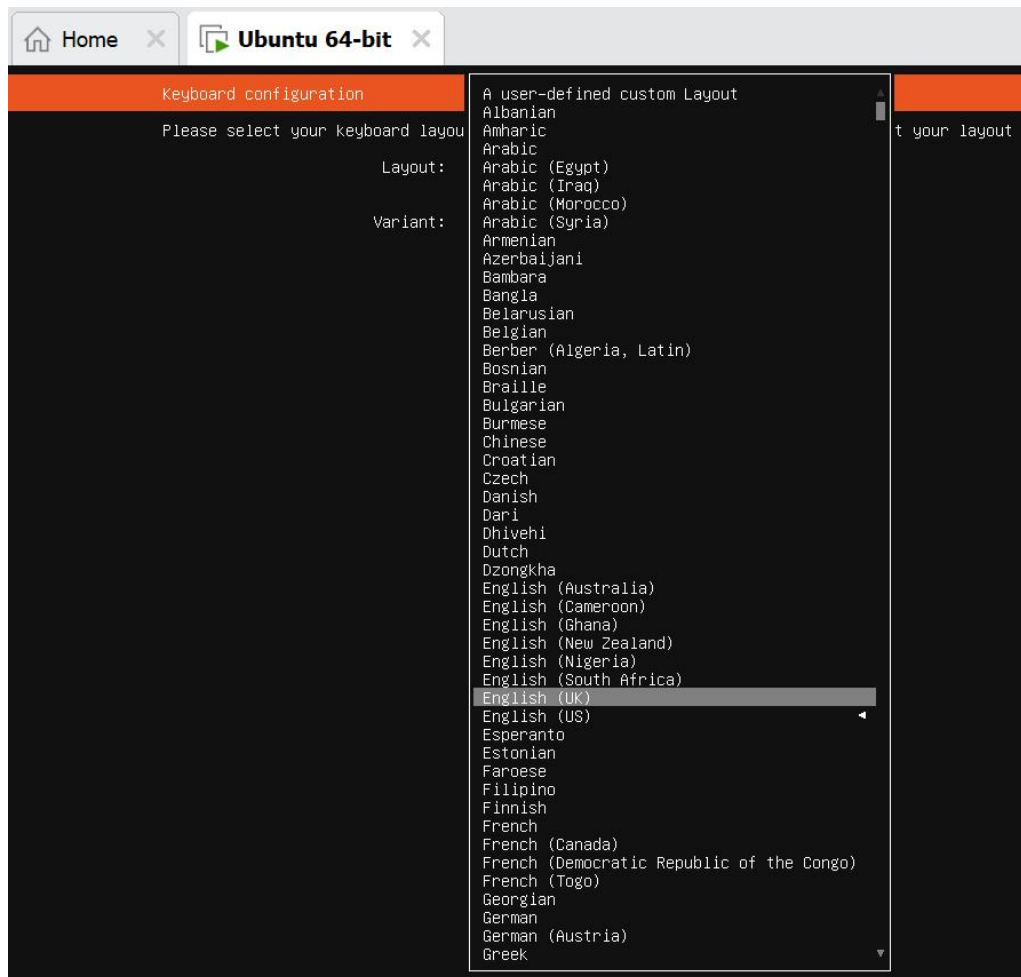
ubuntu_language:



The installer's language selection screen with English chosen as the installation language.

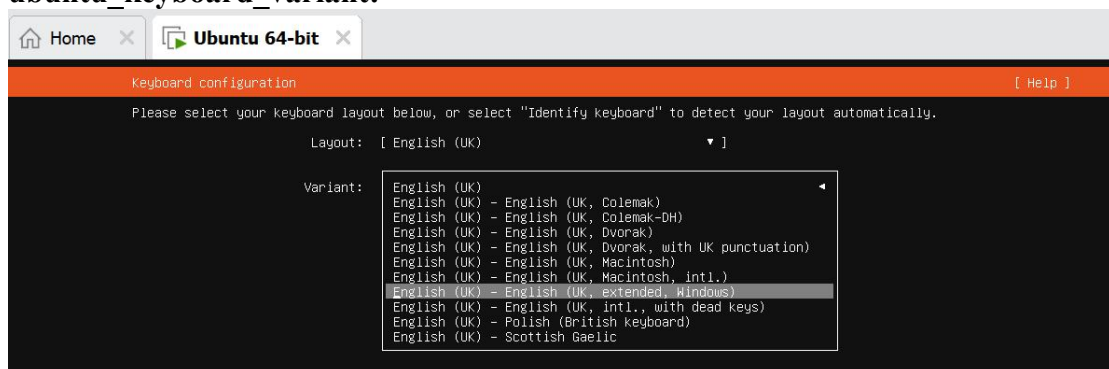
Task 6:

ubuntu_keyboard_layout:



The keyboard configuration screen with English (UK) selected as the keyboard layout.

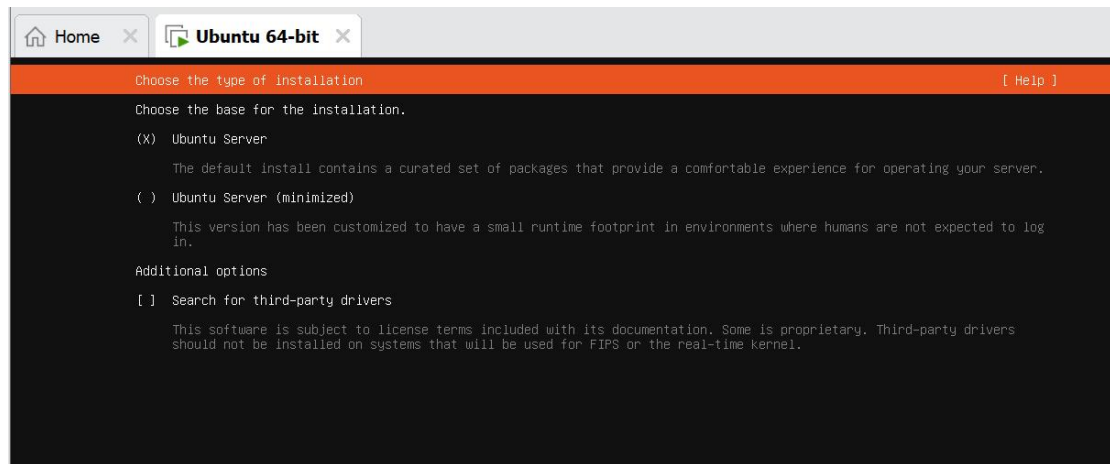
ubuntu_keyboard_variant:



The keyboard variant list for the English (UK) layout, confirming the standard variant selection.

Task 7:

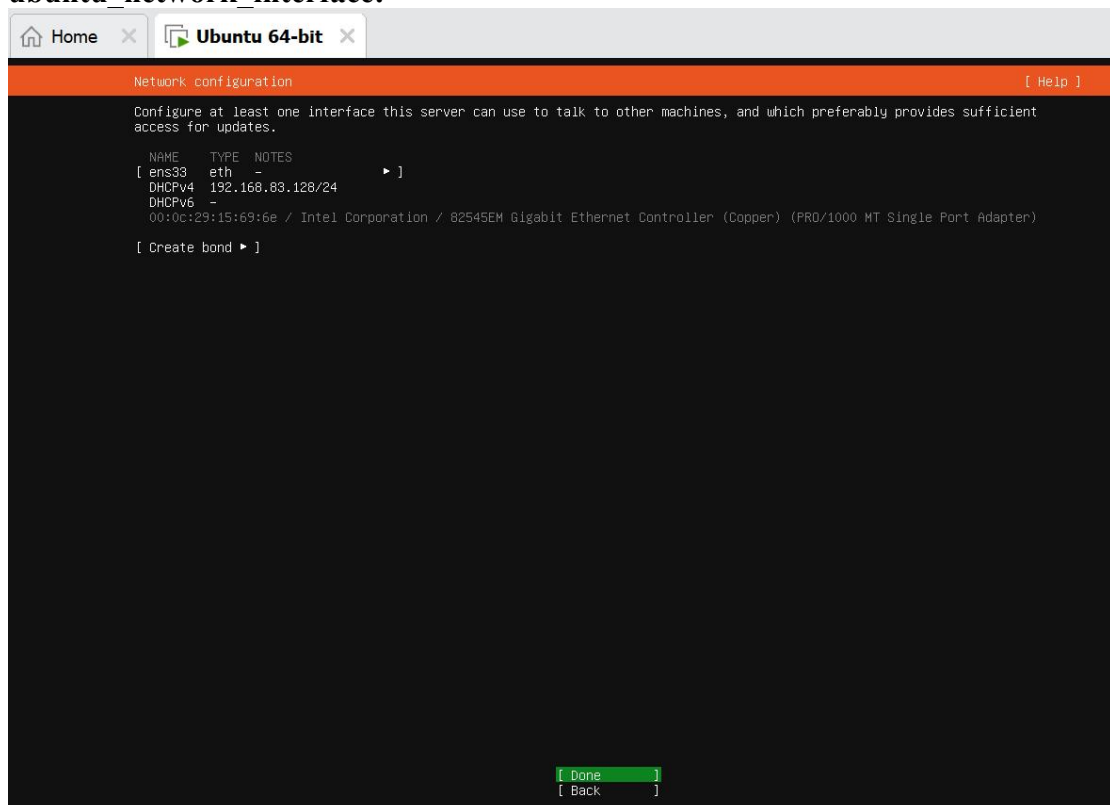
ubuntu_installation_type:



The installation type screen with the full Ubuntu Server base selected rather than the minimised version, and no third-party drivers requested.

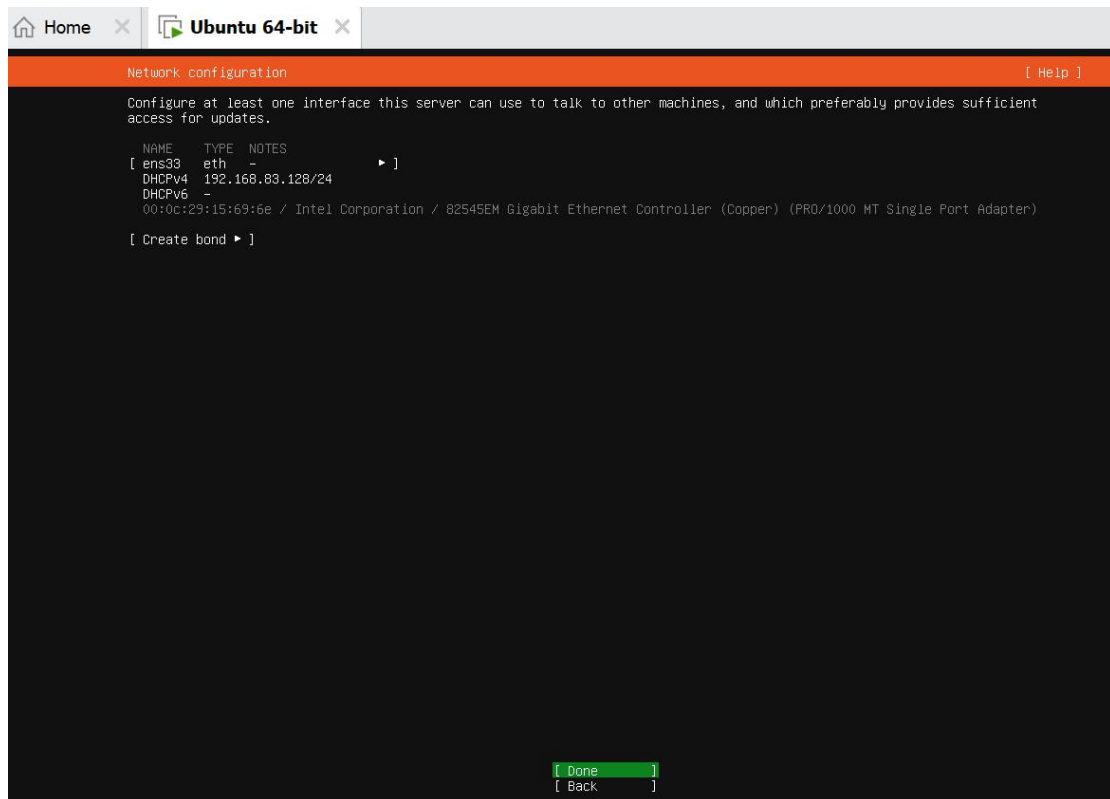
Task 8:

ubuntu_network_interface:



The network configuration screen showing the detected interface ens33 (Intel gigabit adapter) with the DHCP-assigned address 192.168.83.128/24.

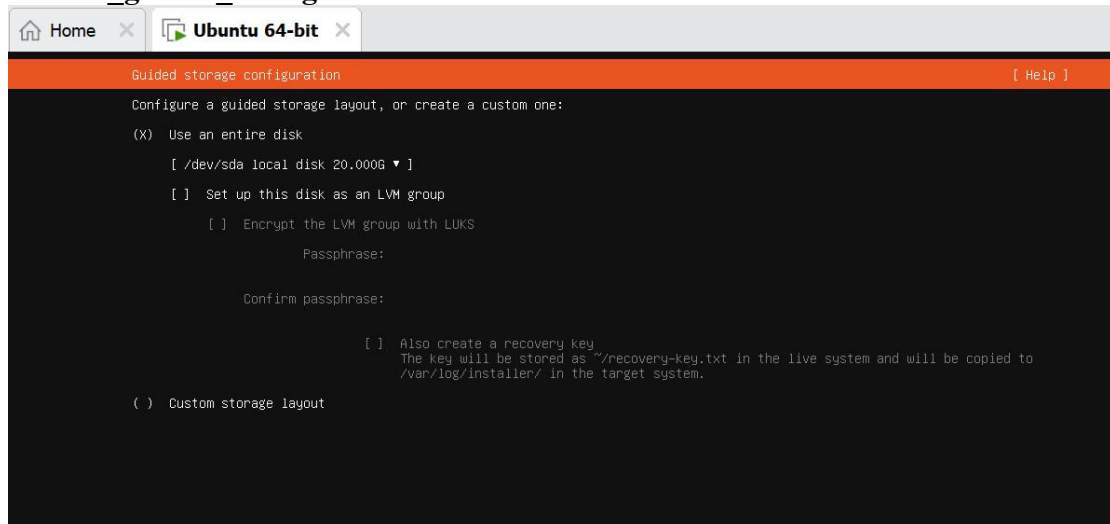
ubuntu_installer_network:



The same network summary accepted with Done, so the VM keeps its automatic DHCP configuration.

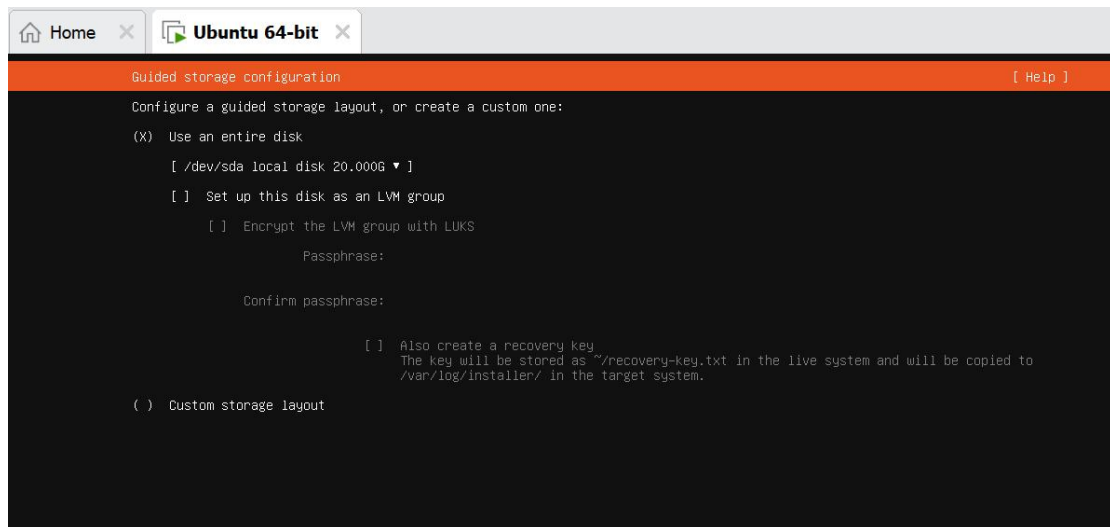
Task 9:

ubuntu_guided_storage:



The guided storage configuration screen with Use an entire disk selected and LVM and encryption left disabled.

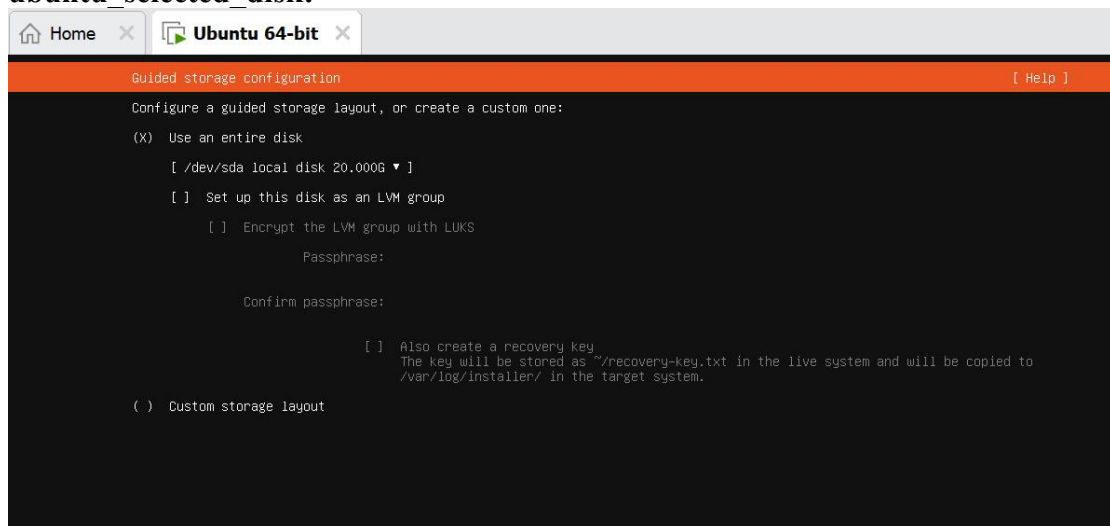
ubuntu_storage_configuration:



The guided storage choices confirmed before continuing to the partition summary.

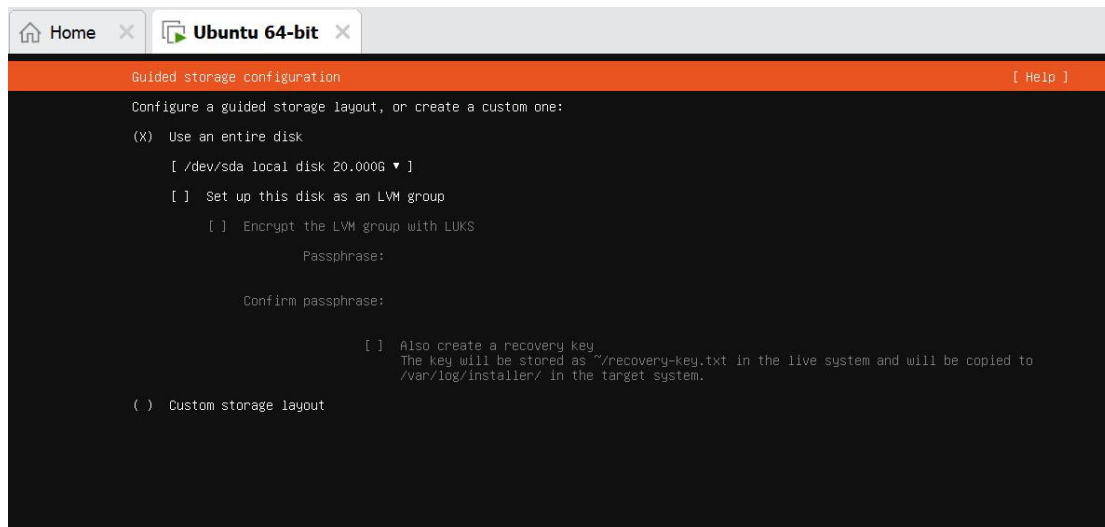
Task 10:

ubuntu_selected_disk:



The target disk chosen for installation: the 20 GB local disk /dev/sda.

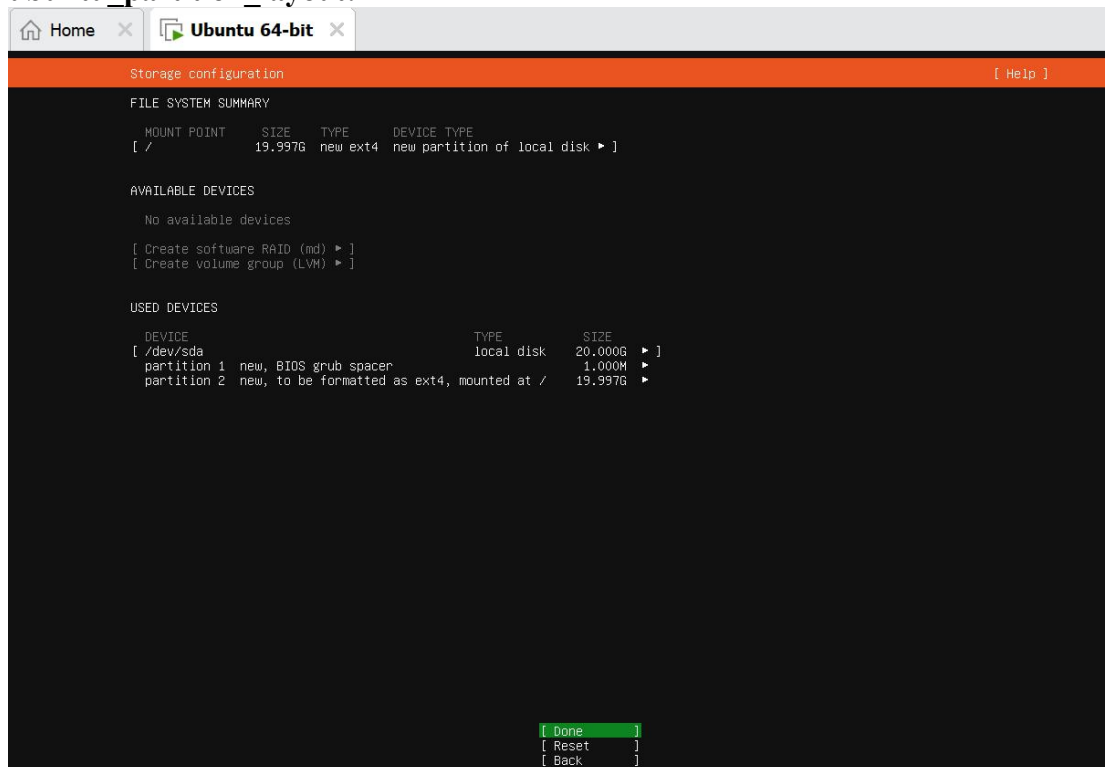
ubuntu_disk_details:



The details of the selected disk, confirming `/dev/sda` is the device the installer will write to.

Task 11:

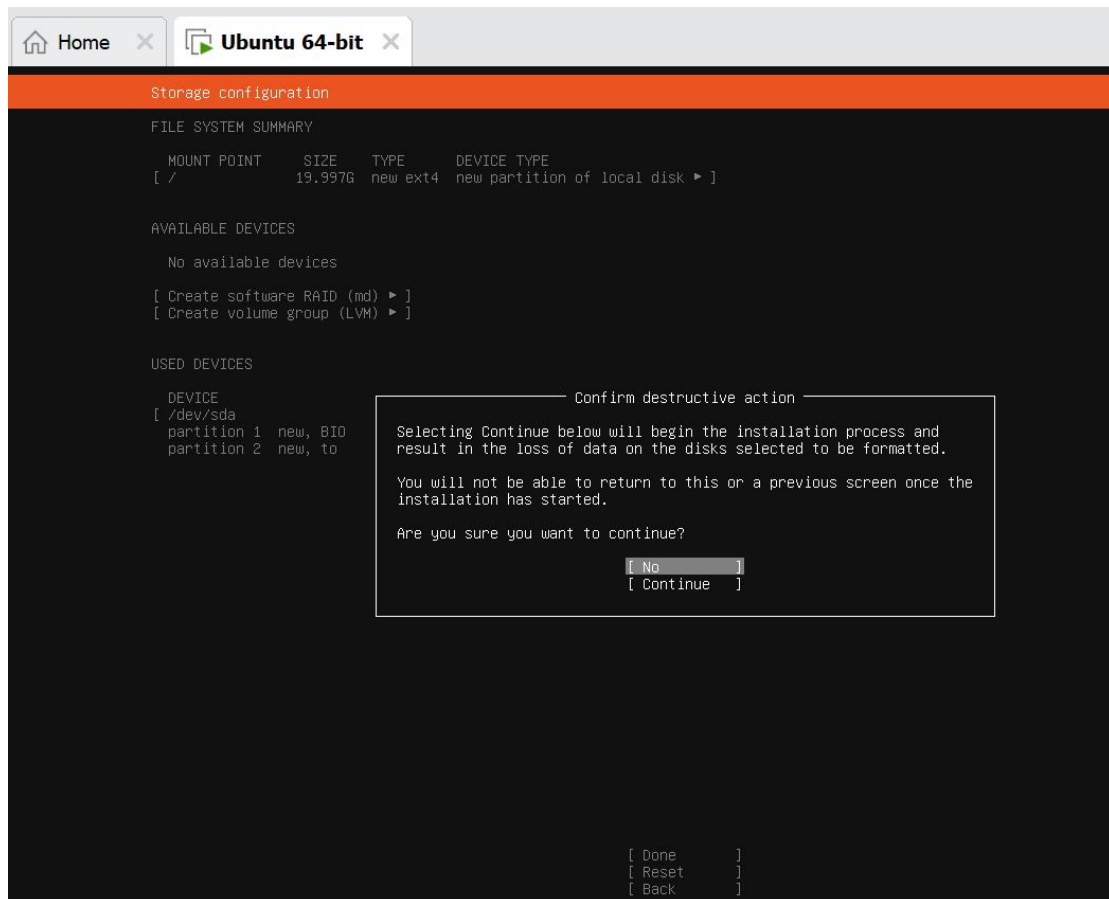
ubuntu_partition_layout:



The resulting partition layout on `/dev/sda`: a 1 MB BIOS grub spacer plus a 19.997 GB ext4 partition mounted at the root (/).

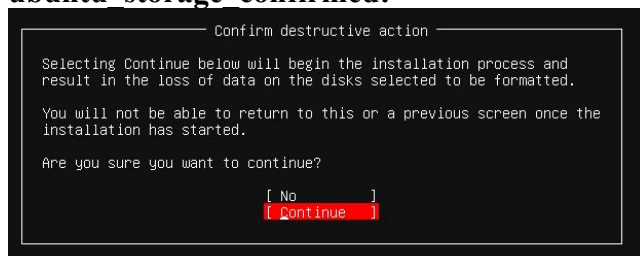
Task 12:

ubuntu_storage_warning:



The installer's Confirm Destructive Action warning, stating that continuing will format the disk and erase existing data.

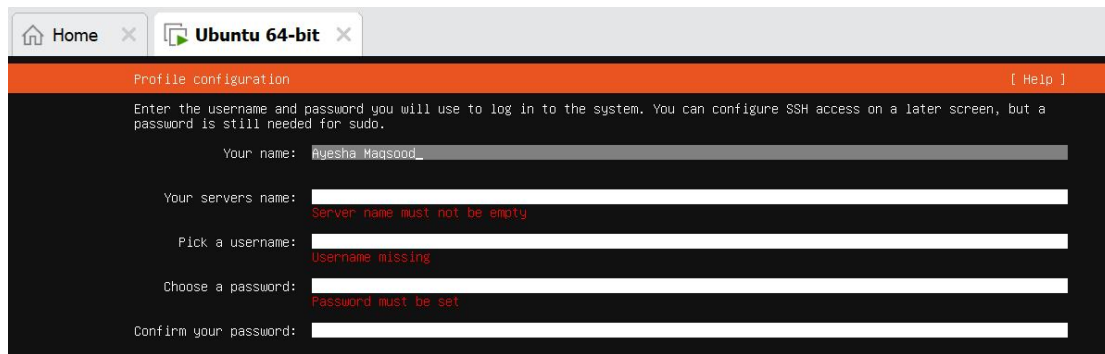
ubuntu_storage_confirmed:



Continue selected on the warning dialog, which commits the storage layout and begins the installation.

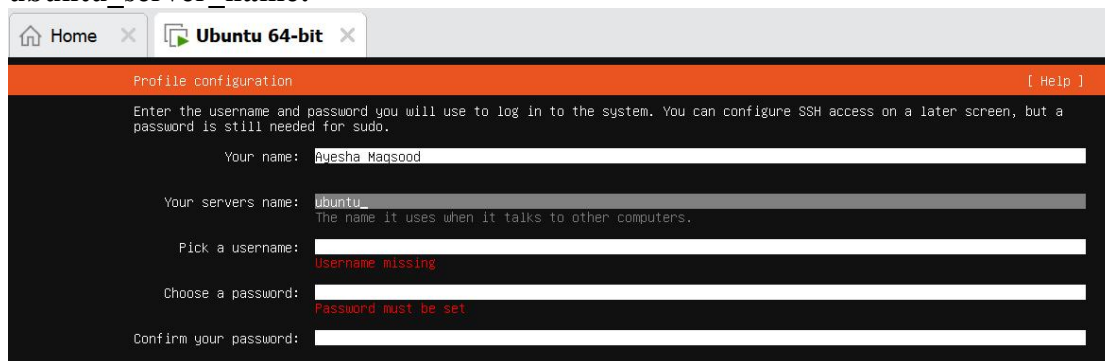
Task 13:

ubuntu_actual_name:



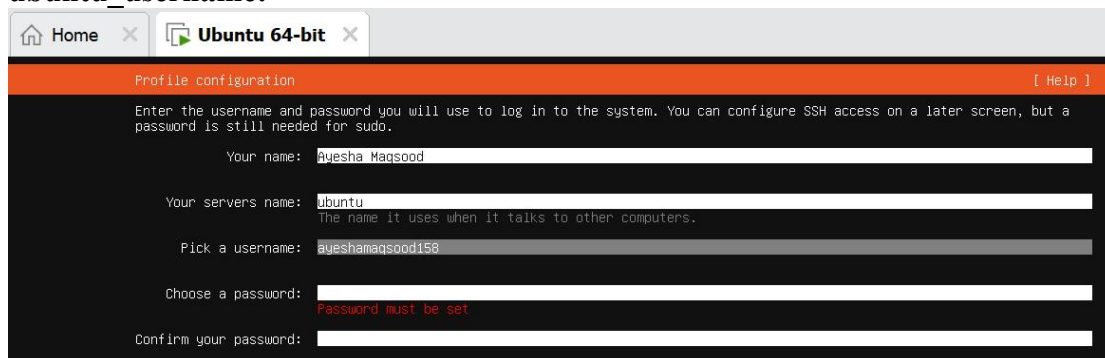
The profile setup screen with the student's real name entered as the account owner.

ubuntu_server_name:



The server hostname set to “ubuntu”, the name the machine uses on the network.

ubuntu_username:



The login username set to ayeshamaqsood158, matching the GitHub username used earlier in the lab.

ubuntu_profile_setup:

Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your servers name:
The name it uses when it talks to other computers.

Pick a username:

Choose a password:

Confirm your password:

[Done]

The completed profile configuration with name, hostname, username and password all set, ready to submit with Done.

Task 14:

ubuntu_installation_progress:

Installing system [Help]

```

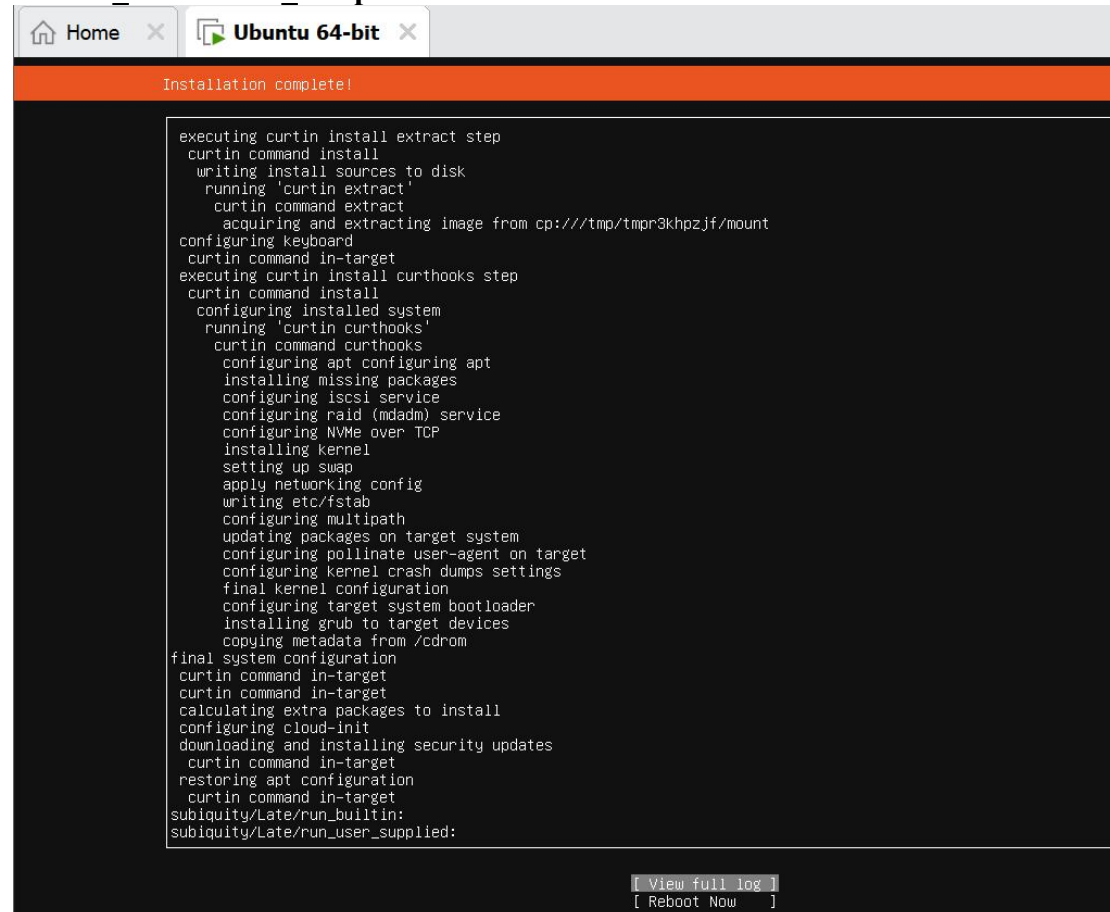
configuring apt
curtin command in-target
installing system
executing curtin install initial step
executing curtin install partitioning step
curtin command install
  configuring storage
    running 'curtin block-meta simple'
    curtin command block-meta
    removing previous storage devices
    configuring disk: disk-sda
    configuring partition: partition-0
    configuring partition: partition-1
    configuring format: format-0
    configuring mount: mount-0
  executing curtin install extract step
  curtin command install
    writing install sources to disk
    running 'curtin extract'
    curtin command extract
    acquiring and extracting image from cp:///tmp/tmpo3khpzjf/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
  configuring installed system
    running 'curtin curthooks'
    curtin command curthooks
    configuring apt configuring apt
    installing missing packages
    configuring iscsi service
    configuring raid (mdadm) service
    configuring NVMe over TCP
    installing kernel
    setting up swap
    apply networking config
    writing etc/fstab
    configuring multipath
    updating packages on target system
    configuring pollinate user-agent on target
    configuring kernel crash dumps settings
    final kernel configuration -
  
```

[View full log]

The installation in progress, with the curtin log showing the system being written to disk and the kernel and bootloader installed.

Task 15:

ubuntu_installation_complete:



```
executing curtin install extract step
curtin command install
  writing install sources to disk
    running 'curtin extract'
    curtin command extract
      acquiring and extracting image from cp:///tmp/tmp3khpzjf/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
  configuring installed system
    running 'curtin curthooks'
    curtin command curthooks
      configuring apt configuring apt
      installing missing packages
      configuring iscsi service
      configuring raid (mdadm) service
      configuring NVMe over TCP
      installing kernel
      setting up swap
      apply networking config
      writing etc/fstab
      configuring multipath
      updating packages on target system
      configuring pollinate user-agent on target
      configuring kernel crash dumps settings
      final kernel configuration
      configuring target system bootloader
      installing grub to target devices
      copying metadata from /cdrom
final system configuration
curtin command in-target
curtin command in-target
  calculating extra packages to install
  configuring cloud-init
  downloading and installing security updates
    curtin command in-target
  restoring apt configuration
    curtin command in-target
subiquity/Late/run_builtin:
subiquity/Late/run_user_supplied:
```

[View full log]
[Reboot Now]

The Installation Complete screen with the Reboot Now option, confirming Ubuntu Server was installed successfully.

ubuntu_login_screen:

```
ubuntu login:
ubuntu login: ayeshamaqsood158
Password:
```

The console login prompt after the reboot, with the username entered at the ubuntu login line.

ubuntu_terminal_login:

```

ubuntu login:
ubuntu login: ayeshamaqsood158
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Sep 15 07:21:18 PM UTC 2026

System load:  0.18           Processes:            215
Usage of /:   28.1% of 19.51GB Users logged in:          0
Memory usage: 21%           IPv4 address for ens33: 192.168.83.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

12 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ayeshamaqsood158@ubuntu:~$ ^C
ayeshamaqsood158@ubuntu:~$

```

The welcome message after a successful first login: Ubuntu 26.04.1 LTS, 28.1% of the 19.51 GB disk used, and the ens33 address 192.168.83.129.

ubuntu identity verified:

```

ayeshamaqsood158@ubuntu:~$ ^C
ayeshamaqsood158@ubuntu:~$

```

The shell prompt ayeshamaqsood158@ubuntu, confirming the logged-in user and hostname on the new server.

Task 16:

ubuntu ip addr command:

```

ayeshamaqsood158@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:15:69:6e brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c2915696e
    inet 192.168.83.129/24 metric 100 brd 192.168.83.255 scope global dynamic ens33
        valid_lft 1065sec preferred_lft 1065sec
    inet6 fe80::20c:29ff:fe15:696e/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
ayeshamaqsood158@ubuntu:~$

```

The output of the ip addr command, listing the loopback interface and ens33 with its MAC address and assigned IPv4 and IPv6 addresses.

ubuntu ip address:

```

inet 192.168.83.129/24 metric 100 b

```

A close-up of the ens33 IPv4 address 192.168.83.129/24, the address used to reach the server over SSH.

Task 17:

windows_ssh_command:

```
PS C:\Users\NEXGEN> ssh ayeshamaqsood158@192.168.83.129
The authenticity of host '192.168.83.129 (192.168.83.129)' can't be established.
ED25519 key fingerprint is SHA256:Fj062SBsG04jAc0XYvtTu9u7xE/IDb5LXRPCpcavg6c.
This key is not known by any other names.
```

The ssh command run from Windows PowerShell to ayeshamaqsood158@192.168.83.129, with the host-authenticity prompt shown on first connection.

windows ssh fingerprint:

```
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.83.129' (ED25519) to the list of known hosts.
ayeshamaqsood158@192.168.83.129's password:
```

The server's ED25519 key fingerprint accepted, adding 192.168.83.129 to the list of known hosts.

windows ssh login:

```
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.83.129' (ED25519) to the list of known hosts.
ayeshamaqsood158@192.168.83.129's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Sep 15 07:45:01 PM UTC 2026

System load:  0.27               Processes:           222
Usage of /:   28.2% of 19.51GB   Users logged in:    0
Memory usage: 24%               IPv4 address for ens33: 192.168.83.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

ayeshamaqsood158@ubuntu:~$
```

The remote session banner after a successful SSH login, showing the same Ubuntu 26.04.1 system and its IP address from the Windows host.

windows ssh identity:

```
ayeshamaqsood158@ubuntu:~$
```

The prompt inside the SSH session, confirming the remote user and hostname and that the server can be administered remotely from Windows.